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3.1.3 What you will learn

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This week focuses on the QR factorization as a method for computing an orthonormal basis for the column space of a matrix.

Upon completion of this week, you should be able to

- Relate Gram-Schmidt orthogonalization of vectors to the QR factorization of a matrix.
- Show that Classical Gram-Schmidt and Modified Gram-Schmidt yield the same result (in exact arithmetic).
- Compare and contrast the Classical Gram-Schmidt and Modified Gram-Schmidt methods with regard to cost and robustness in the presence of roundoff error.
- Derive and explain the Householder transformations (reflections).
- Decompose a matrix to its QR factorization via the application of Householder transformations.
- Analyze the cost of the Householder QR factorization algorithm.
- Explain why Householder QR factorization yields a matrix Q with high quality orthonormal columns, even in the presence of roundoff error.

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